The Wrong Unit of Uncertainty: Simultaneous Conformal Bands for Repeated Cross-National Surveys

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Abstract

Comparative research routinely reads a single wave-pair contrast as a persistent, distribution-wide trend—attaching uncertainty to the wrong unit and over-detecting change. We give a deployable instrument: a finite-sample simultaneous band over a country's whole attitude *trajectory*, with the country as the exchangeable unit and an ordered hierarchy of claims—exact at any K for the estimated trajectory, explicitly bounded for the latent one, and inside a selector whose nested fallbacks make the guarantee hold under *any* selection rule. In the ESS the hierarchy cuts a marginal reading of twenty of thirty countries to net decline in six. Over the full 2002–2024 record none certifies a persistent slide; reading every ordered contrast and both directions off one band, twenty-three of thirty-three countries certify both a fall and a rise, and span erosion certifies in eight. On the World Values Survey the same shift of rung cuts the certified set 1.9–4.8× at fixed inference, and what survives is post-communist and Arab-Spring states. The second error—calibration objects estimated from complex samples—we resolve by scope: that correction is non-identified and unreachable at the national unit, though both of its barriers are reachable one unit down.

1 Introduction

Comparative politics increasingly treats a national public as a *distribution* rather than a mean. Trust in parliament in a country and year is a cumulative distribution function over a response scale, and repeated cross-national surveys—the European Social Survey (ESS; Kaminska and Lynn, 2017), the World Values Survey, the AmericasBarometer (Cohen et al., 2021)—let us track how that distribution moves across waves and *transport* it: predict one country's trajectory from the others, or flag departures from a common pattern. Conformal methods are a natural tool, promising finite-sample, distribution-free coverage.

Two wrong units. A claim and its calibration can each be attached to the wrong object. The *claim* "democratic trust is declining" is typically built from a single wave-pair mean contrast, then narrated as a country-level trend. Our instrument is a finite-sample simultaneous band over a country's whole trajectory, with the country as the exchangeable unit, and an ordered hierarchy of claims (pairwise $\to$ any-pair $\to$ net $\to$ persistent $\to$ Bonferroni); the top rung is *strictly stronger* than the net-trend claim the literature typically intends, and we treat the two as co-headlines. This multiplicity-honest core needs no survey design and applies to any repeated distributional claim.

The second, subtler wrong unit is the calibration: the objects conformal transport calibrates on—other countries' attitude distributions—are themselves *estimated* from complex probability samples. Standard practice plugs these estimates in as truth, blind to each "country" being a noisy estimate of a latent population function.

The two errors are distinct. Writing $F_{c,r}(t)$ for country c's latent CDF at threshold t in wave r and $\widetilde{F}_{c,r}(t)$ for its complex-survey estimate,

$$\widetilde{F}_{c,r}(t) - \mu_r(t) \;=\; \underbrace{\big(F_{c,r}(t) - \mu_r(t)\big)}_{\text{transport error}} \;+\; \underbrace{\big(\widetilde{F}_{c,r}(t) - F_{c,r}(t)\big)}_{\text{design error}}, \tag{1}$$

where μ_r is the transport center. A band calibrated on the observed $\widetilde{F}$ scores treats the sum as the transport error alone: when the design error is non-negligible the band is too wide, yet any correction subtracting the survey noise can make it too narrow, being itself estimated from few countries.

Contributions. We make four. *First, a diagnosis and a portable instrument.* We separate the claim unit (a wave-pair contrast standing in for a distribution-wide trajectory) from the calibration unit (the omitted design-error layer of (1)), and give a finite-sample simultaneous band over a country's whole trajectory with a multiplicity-honest guarantee hierarchy that corrects the first. *Second, a provably-valid deployed procedure.* The band we recommend is a clustered "population" conformal band (PCB), finite-sample exact at *any* K (Theorem 3), inside a selector whose validity does not depend on how it selects: the PCB and conservative branches are *nested*, and selection among nested bands anchored at an exact band is validity-free, so the four target-blind gates are an efficiency device, not a validity assumption (Theorem 5). At cross-national K the deployed guarantee is exact $1-\alpha$ for the estimated trajectory, with a design-noise-bounded gap to the latent one. We validate the frozen procedure in three disclosed layers, ending with a fresh sealed run on six unseen DGP families. *Third, two named reanalyses* (Section 8) where the instrument changes the substantive picture. For European trust the hierarchy certifies net decline in six of thirty (2018–2024), not the twenty flagged marginally; over the full 2002–2024 record *no* country certifies persistence, twenty-three of thirty-three certify both declines and recoveries, and eight certify span erosion, several of them with few certified adjacent contrasts—erosion an adjacent-wave reading would miss. All of these are read off one band at one level, by a claim-family transfer argument that removes the multiplicity and endpoint-selection corrections a per-rung construction would need. Globally, a wave-pair reading of the Foa–Mounk "deconsolidation" battery on the WVS/EVS (1981–2022) over-counts the persistent phenomenon by 2.6–6.5×; aggregating per-item certifications turns the correction into a discovery—the thirteen-country *certified core* concentrates in post-communist and Arab-Spring-aftermath states, with no consolidated Western democracy certifying on more than one item. *Fourth, the scope condition that justifies this deployment, and where it ends.* Without the design-noise law the latent transport-error distribution is non-identified (Theorem 1); with it, the correction is still *unreachable* at cross-national scale, because the measured design-to-total ratio stays severalfold below the level at which deconvolution helps while the reliability diagnostic obeys the algorithmic floor $D \geq \sqrt{2/(K-1)}$, requiring $K \geq 94$ exchangeable populations. The two largest surveys fail in *opposite* ways: the WVS clears the size barrier ($K \approx 100$) with negligible weights-only noise, the ESS has genuine clustered noise but $K \leq 33$. Both failures are properties of the *unit*: refining to ESS subnational regions carries K to 300 and lifts ρ to the gate, so both barriers are reachable on real survey data and the selector then abstains on efficiency alone (Section 7). Refinement is not free—with unit size controlled, narrower units strictly widen the band—and the one hard constraint on the calibration unit is a feasibility floor, $K \geq \alpha^{-1} - 1$, below which the conformal level cannot be met at all.

The remainder situates the contribution (Section 2), sets up the problem and the impossibility result (Section 3), builds the method and its theory (Sections 4–5), validates it (Section 6), characterizes the real-data regime (Section 7), reanalyzes political trust (Section 8), and discusses scope (Section 9).

2 Related work

Four literatures intersect here; the Supplementary Material gives the extended discussion. *Conformal prediction* (Vovk et al., 2005; Lei et al., 2018; Barber et al., 2023): our selector adapts not to temporal shift (Gibbs and Candès, 2021) but to the certified magnitude of survey design noise; within the discipline, Randahl et al. (2026) calibrate bin-conditional forecast intervals, a different object from simultaneous trajectory bands. *Cluster-as-unit conformal* (Dunn et al., 2023; Lee et al., 2023) transports guarantees to an unseen cluster but treats each cluster's data as cleanly sampled (the same clustered construction appears, for a non-survey domain, in a companion paper by the present authors; reference omitted for review); our departure is that the calibration objects are *estimated* under complex designs (Binder, 1983; Lumley, 2010). *Noisy calibration*: Wieczorek (2023) already brings conformal prediction to complex survey designs, for unit-level prediction under design-based inference—our departure is the simultaneous trajectory object, the population as the exchangeable unit, and the estimated-calibration boundary; Einbinder et al. (2024); Sesia et al. (2025) deconvolve a *known* label-noise model, an identification premise that generally fails for design noise (Theorem 1), and even a supplied noise law faces the finite-K floor (Proposition 1). *Functional simultaneous bands* (Diquigiovanni et al., 2021): unlike FDR/Bonferroni corrections of marginal tests (Benjamini and Hochberg, 1995), the "persistent" object is covered directly by one band—a distinction of estimand, not error rate (Gelman et al., 2012). Substantively we engage the trust-decline and deconsolidation debates (Norris, 2017; Voeten, 2017; Wuttke et al., 2022; Valgarðsson et al., 2025; Foa and Mounk, 2016, 2017; Ouattara and van der Meer, 2023; van der Meer and Janssen, 2025; Kołczyńska and Bürkner, 2026); the closest substantive competitor is Claassen (2019, 2020), whose latent-variable panels model measurement and sampling uncertainty with model-based credible intervals—intervals whose calibration is itself contested (Tai et al., 2024)—our bands trade pooling and width for finite-sample, model-free coverage at a stated claim rung, and the two are complements. The certified core's geography intersects the post-communist disillusionment literature (Mishler and Rose, 1997; Inglehart and Catterberg, 2003), which anticipated the decline-from-euphoric-baselines pattern our certification formalizes per country.

3 Setup and the two targets

The object we deploy is the exact simultaneous band of Theorem 3; the impossibility below (Theorem 1) is a scope boundary delimiting when a design-aware correction to that band is even available.

Objects. We observe K source countries, indexed $c = 1, \dots, K$, treated as exchangeable units (the "population" or cluster is the exchangeable object, not the individual respondent). Each country has a latent attitude trajectory $F_{c,r}(t)$, a CDF over a fixed grid of thresholds $t \in \{t_1, \dots, t_T\}$ and survey rounds r. We do not see $F_{c,r}$; we see a complex-survey estimate $\widetilde{F}_{c,r}(t) = F_{c,r}(t) + S_{c,r}(t)$, where $S_{c,r}$ is design (sampling) error with a design variance $v_{c,r}(t)^2$ that a stratified-PSU / replicate-weight bootstrap can estimate. Define the transport score of a country as its worst-case standardized deviation from the transport center,

$$R_c \;=\; \max_{r,t} \frac{|F_{c,r}(t) - \mu_r(t)|}{\sigma(t)}, \qquad \widetilde{R}_c \;=\; R_c + \xi_c,$$

where $\widetilde{R}_c$ is the same quantity computed from the observed $\widetilde{F}$ and ξ_c is the induced design contamination. (Because R is a supremum, ξ_c is not exactly mean-zero or independent of R_c: the additive centred score model is a working approximation, exact at the curve scale under (A1), whose failure direction is conservative for the plug-in band.) The governing scalar is the design-to-*total* SD ratio

$$\rho \;=\; \frac{\mathrm{SD}(\xi)}{\mathrm{SD}(\widetilde{R})} \;=\; \sqrt{\frac{\overline{v^2}}{s_R^2 + \overline{v^2}}} \;\in\; [0, 1),$$

$\rho \to 0$ the design-noise-negligible regime, $\rho \to 1$ noise-dominated. (The simulations' generative dial is the unbounded $\mathrm{SD}(\xi)/\mathrm{SD}(R)$; reported ρ values use the bounded definition.)

Target and validity. We want a simultaneous band covering a *new* country's *latent* trajectory, $\Pr\left(F_{\mathrm{new},r}(t) \in \widehat{B}(r,t)\ \forall r, t\right) \geq 1-\alpha$. Two facts frame everything. Ordinary clustered conformal on the observed scores is exact for the *survey-estimate* target $\widetilde{F}_{\mathrm{new}}$, needing only exchangeability; its coverage of the *latent* target differs by an explicit gap that vanishes with the design-noise scale (Theorem 2(b); symmetry of the design law alone is *not* enough for conservativeness). Meanwhile its width is inflated by design noise by a factor growing with ρ, and correcting that requires removing the design contribution—a deconvolution.

Country exchangeability, stated honestly. The transport band's identifying assumption is a superpopulation postulate: participating countries' trajectory laws are symmetric draws—over a set that is not sampled, enters and exits rounds non-randomly, and changes designs across rounds. A strong idealization, tempered three ways: the headline certifications of Section 8 do not use it (they are within-country design-bootstrap statements); under approximate exchangeability the Barber et al. (2023) bound controls the coverage gap by an unobserved total-variation distance, treated as qualitative; a leave-one-region-out audit finds near-nominal held-out coverage except for the post-communist East (6/9 and 5/9 at nominal 0.90)—a quantified scope condition on transport *into* that bloc (Supplementary Material); and the Bonferroni rung protects against screening multiplicity—a distinct safeguard that does not repair non-exchangeability.

The impossibility. The central obstruction is that the deconvolution is not identified from the contaminated scores alone.

Theorem 1 (Non-identification and necessity of design information)**.** *Let the observed score be $\widetilde{R} = R + \xi$ with $R \perp \xi$. (i) The law of R is not identified from the law of $\widetilde{R}$ without restrictions on the law of ξ: there exist distinct pairs $(\mathcal{L}_R, \mathcal{L}_\xi) \neq (\mathcal{L}'_R, \mathcal{L}'_\xi)$ with $\mathcal{L}_R * \mathcal{L}_\xi = \mathcal{L}_{R'} * \mathcal{L}_{\xi'}$. (ii) Consequently, among non-randomized bands that are measurable functions of the observed scores only, any band whose radius lies almost surely below the $(\lceil(1-\alpha)(K+1)\rceil-1)$-th order statistic of $\{|\widetilde{R}_c|\}$ has latent coverage at most $(\lceil(1-\alpha)(K+1)\rceil-1)/(K+1) < 1-\alpha$ on the member $\xi \equiv 0$ of the class, which it cannot distinguish from the contaminated one. The only room below the plug-in radius is the conformal discreteness slack, which randomized smoothing exhausts;* uniform *improvement beyond it requires shrinking the class of admissible noise laws.*

The reading is constructive (proof in Appendix A): a band both honest and narrower than the plug-in must *supply* the design-noise law—replicate weights or a design bootstrap, exactly what Section 4 uses.

Remark 1 (Beyond surveys). Nothing here is survey-specific: the result holds whenever conformal calibration uses *estimated* objects—MRP small-area outputs, learned or imputed features, meta-analytic effects. Design/measurement information is the identifying ingredient, and its reliability at finite K is itself bounded (Proposition 1); confirmed outside surveys on a small-area calibration (Supplementary Material).

4 A nominal-safe adaptive method

Theorem 1 says the design bootstrap is the identifying input. The method turns that input into a band by choosing, target-blind, among three constructions and by never using the aggressive one unless it is provably reliable at the available number of countries.

Three branches. For calibration error curves $\{\widetilde{E}_c\}_{c=1}^{K}$ over threshold grid t, with per-country design SD $\widehat{v}_c(t)$ from the stratified-PSU bootstrap: *(i) Clustered PCB* takes the conformal quantile of the *unstudentized* scores $\max_t |\widetilde{E}_c(t)|$ (the U0 construction): exact for the survey-estimate target at any K, with the latent gap of Theorem 2(b). *(ii) Deconvolution* studentizes by the finite-K-safe target scale $\widehat{s}_T^2(t) = \max\bigl(s^2_{\text{plug}}(t) - [\,\overline{\widehat{v}^2}(t) - z\ \text{SE}\,]_+,\ \text{floor}\bigr)$, which subtracts a *lower* confidence bound on the design variance so the correction cannot over-shrink; its width scales as $\sqrt{1-\rho^2}$ relative to PCB. *(iii) Conservative* inflates each unstudentized score by the design SD, $\max_t(|\widetilde{E}_c(t)| + z\,\widehat{v}_c(t))$ — a score that dominates the PCB score pointwise, so the conservative band *contains* the PCB band by construction. This nesting is load-bearing: selection between nested bands anchored at an exact band is validity-free, so the selector needs no conditional-coverage argument.

The three questions. A naive selector asks *can we deconvolve* and *do we need to*; the safe selector adds the missing third question, *can we do it safely at this K?*

Safe selector (Algorithm 1). Let $\widehat{\rho}_{\mathrm{LCB}}$ be a conservative lower bound on ρ, $D = \max_t \mathrm{SE}(\widehat{s}_T^2)/\widehat{s}_T^2$ a reliability diagnostic, and $\widehat{\delta}_{\mathrm{UCB}}(D)$ a frozen upper bound on the finite-K coverage remainder. Deconvolution is used *iff all four gates pass*: (A) $\widehat{\rho}_{\mathrm{LCB}} > \rho_0$ (design noise materially large); (B) $\widehat{\delta}_{\mathrm{UCB}}(D) \leq \delta_{\max}$ (remainder within budget); (C) $\widehat{\mathrm{Gain}} - \Delta_g > g_{\min}$ (beats conservative by a margin); (D) stability. Otherwise the band is PCB when $\widehat{\rho}_{\mathrm{LCB}} \leq \rho_0$ and the conservative envelope when design noise is real but information insufficient. Every gate is a function of the calibration set only, so the selector is target-blind—and by Theorem 5's nested-band argument, validity survives even adversarial selection between the anchor branches. The deconvolution branch, which is *not* nested, is charged its own miss budget $\alpha_{\mathrm{dec}} = \min\{\max(0.1\alpha, 3/(K+1)), \alpha/2\}$ (the anchors run at $\alpha - \alpha_{\mathrm{dec}}$), and only in the $K \geq 94$ regime where its gate can open at all; at the K of every certification sample in this paper the anchors keep the full α and the deployed guarantee is exact. All constants are frozen before validation (Section 5, Appendix B); when deconvolution is used the reported level is the observable $1 - \alpha - \widehat{\delta}_{\mathrm{UCB}}(D)$, a frozen *calibrated* bound validated in Section 6 (the theorem-level guarantee is Theorem 4's $\varepsilon_{K,B}$).

Deployment. The method ships as one call, `dapcb(cal_errors, v_cal, center, alpha)`, in an installable Python package (`pcb`), returning the band, selected branch, diagnostics, coverage level, and fallback reason—the user sees why the band was built and reads off its finite-K guarantee.

5 Theory

We state validity for the oracle, the estimated-law procedure, and the deployed selector, then efficiency. Proofs are in Appendix A; each theorem has a machine-checked contract test.

Algorithm 1: Nominal-safe adaptive design-aware band

Input: calibration curves $\{\widetilde{R}_c\}$ (unit = country), design replicates $\{\widehat{v}_c\}$, target curve $\widehat{F}_{\text{new}}$, level α; frozen $\rho_0, \delta_{\max}, g_{\min}, \Delta_g$ and the K-only budget rule $\alpha_{\text{dec}}(K)$

set $(\alpha_{\text{anchor}}, \alpha_{\text{dec}})$ by the frozen budget rule (full α to the anchors when gate B is infeasible at this K);

compute diagnostics s, $\widehat{s}_T$, $\widehat{\rho}_{\text{LCB}}, D, \widehat{\delta}_{\text{UCB}}(D)$; the nested unstudentized anchor radii $r_{\text{PCB}}, r_{\text{con}}$ at level α_{anchor}; and r_{dec} at level α_{dec};

if $\widehat{\rho}_{LCB} \le \rho_0$ **then**
  branch $\leftarrow$ PCB;
else if $\widehat{\delta}_{UCB}(D) \le \delta_{\max}$ ***and*** *stable* ***and*** $\widehat{Gain} - \Delta_g > g_{\min}$ **then**
  branch $\leftarrow$ deconvolution;
else
  branch $\leftarrow$ conservative;

return $\widehat{F}_{\text{new}} \pm r_{\text{branch}}$, branch, and diagnostics;

Theorem 2 (Oracle validity)**.** *Suppose the countries are exchangeable and the design law is known.* (a) *The clustered band is finite-sample valid for the survey-estimate target:* $\Pr(\widetilde{F}_{new} \in \widehat{B}) \ge 1-\alpha$*, using exchangeability only.* (b) *For the latent target the same band satisfies, for every* $u > 0$, $\Pr(F_{new} \in \widehat{B}) \ge 1 - \alpha - \Pr(\xi < -u) - \Pr(\widetilde{R} \in (\widehat{q}-u, \widehat{q}])$*: a noise-tail term plus a local anti-concentration term, vanishing with the design-noise scale (*$\rho^{2/3}$ *under variance-only tails,* ρ *up to logarithms under sub-Gaussian noise). Symmetric design noise alone does* not *make the band conservative for the latent target—a bimodal counterexample is in the Supplementary Material—which is why we report the observed* ρ. (c) *The deconvolution band with the true target scale is valid for the latent target—finite-sample* exact *under the scale-family assumption (A1) of Theorem 4—with width a factor* $\sqrt{1-\rho^2} + o(1)$ *of the clustered band.*

Theorem 3 (Exact finite-K validity of the deployed base band)**.** *Let the base band use the* unstudentized *cluster score* $R_c = \max_t |E_c(t)|$ *(no data-dependent modulation), with a center satisfying the symmetric-construction condition: fixed, split-fold, per-unit from that unit's own data (the deployed own-previous-round centers), or a symmetric function of all* $K{+}1$ *curves (Appendix A states the four cases). If the countries are exchangeable, then for* every

K the band $\widehat{B} = center \pm \widehat{q}$, with $\widehat{q}$ the $\lceil(1-\alpha)(K{+}1)\rceil$-th order statistic of $\{R_c\}$, satisfies $\Pr(\widetilde{F}_{new} \in \widehat{B}) \geq \lceil(1-\alpha)(K{+}1)\rceil/(K{+}1) \geq 1-\alpha$, distribution-free—exact for the survey-estimate *trajectory (T1); the latent trajectory (T2) adds the explicit gap of Theorem 2(b), bounded through the reported ρ. Two asymmetries are excluded by construction—in-sample modulation (an $O(g/K)$ self-inclusion deficit) and grand-mean centers containing their own unit (an anti-conservative $O(1/K)$ bias, removed by leave-one-out centers)—and on the near-homoskedastic real curves the width cost of dropping the modulation is $\leq 5\%$.*

Theorem 4 (Estimated-law validity)**.** *Assume a common scale family indexed by the design variance (A1), anti-concentration of the sup-score law near its upper quantile (A2), and bounded curves with an unbiased size-B design bootstrap (A3); the appendix states all three precisely. Then* (i) *with oracle scales the deconvolution band is finite-sample* exact *for the latent target at every K, and* (ii) *with estimated scales it satisfies $\Pr(F_{new} \in \widehat{B}) \geq 1-\alpha-\varepsilon_{K,B}$ with $\varepsilon_{K,B} = O\big(\sqrt{\log(KT)/\min(K,B)}\,\big)$, proved by a sandwich and concentration argument with constants depending only on the constants of (A1)–(A3). The finite-K-safe scale $\widehat{s}^2_T$ (Section 4) subtracts a lower confidence bound on the design variance, making the scale-error term one-sided (wide, not narrow) outside an α_2-probability event. Some shape restriction is not removable—the bimodal counterexample of Theorem 2(b) defeats any latent-target band without one—and (A1) is one sufficient choice, strong enough to make the distributional matching exact.*

Theorem 5 (Safe-adaptive finite-K validity—the deployed pipeline)**.** *Deploy Algorithm 1 with unstudentized PCB and conservative branches (scores $\max_t |E_c(t)|$ and $\max_t(|E_c(t)| + z\widehat{v}_c(t))$, both at level α_{anchor}) and the deconvolution branch at its own budget α_{dec}, where $(\alpha_{anchor}, \alpha_{dec}) = (\alpha, 0)$ when the reliability gate is infeasible at the observed K (Proposition 1: $K < 94$, which covers every certification sample analyzed in this paper, including the WVS ≥ 2-wave sets at $K = 59$–77) and otherwise $\alpha_{dec} = \min\{\max(0.1\alpha,\ 3/(K{+}1)),\ \alpha/2\}$ with $\alpha_{anchor} = \alpha - \alpha_{dec}$, a frozen function of K alone. Then, finite-sample and with* no conditions on the selection rule,

$$\Pr\left(\widetilde{F}_{new,r}(t) \in \widehat{B}\right) \geq 1-\alpha \quad (K<94), \qquad \Pr\left(F_{new,r}(t) \in \widehat{B}\right) \geq 1-\alpha-\varepsilon_{K,B} \quad (K \geq 94).$$

The clauses name different targets because they rest on different assumptions: exchangeability alone gives the estimated trajectory T1 exactly (the latent T2 adding Theorem 2(b)'s gap), while (A1)–(A3)—already inherited from Theorem 4 through its remainder $\varepsilon_{K,B}$, which is attached to the deconvolution branch only—additionally deliver the latent target directly, by an anchor-domination lemma (Appendix A).

The proof uses no conditional-on-selection step: a nested-band lemma (the conservative score dominates the PCB score pointwise, and selection between nested bands anchored at an exact band is validity-free—an instance of nested-conformal machinery (Gupta et al., 2022; Yang and Kuchibhotla, 2024), our contribution being to engineer the branches so it applies) plus a union bound charging the non-nested branch its own budget where its gate can open. Two scope notes: at $K \geq 94$ the *reported* level $1-\alpha-\widehat{\delta}_{\mathrm{UCB}}(D)$ is a frozen calibrated bound, not a theorem; and the unsurveyed-target mode with leave-one-out centers carries a residual $O(1/K^2)$ asymmetry bounded only heuristically, so a strict guarantee there wants a split-fold center.

Efficiency. *(AW-1, proved)* the low-ρ deconvolution/PCB width ratio is $1-\frac{1}{2}\rho^2+O(\rho^4)$; *(AW-2)* conservative $\supseteq$ PCB holds pointwise, its Bonferroni domination resting on an unproved expansion; *(AW-3, proved)* selection cannot inflate miscoverage (Theorem 5). AW-4 (open) and the fitted $O(g/K)$ modulation deficit are in Appendix A.

6 Simulation and a design-preserving stress test

Simulation establishes two facts: attaching the band to the wrong unit collapses coverage, and the safe selector meets its guarantees on data it has never seen.

Table 1: Whole-trajectory coverage (4000 reps, nominal 90%) as trajectory length L grows.

L (rounds)	marginal (per point)	per-round	trajectory (ours)
2	38.2%	81.7%	**90.0%**
4	16.1%	68.2%	**90.2%**
6	7.1%	58.2%	**90.3%**
8	3.5%	49.8%	**90.9%**

6.1 The wrong unit collapses coverage

On a ground-truth simulation ($K = 30$ countries, $T = 6$ thresholds, errors correlated across rounds and thresholds; Table 1) we ask each band for *trajectory* coverage—the whole curve-path at once. All three bands are unstudentized and finite-sample, so only the score's unit varies. The per-round band, marginally valid at the round level, covers the whole trajectory only 0.82 at $L = 2$ and 0.50 at $L = 8$; the marginal band collapses faster; the country-trajectory band holds ≈ 0.90 at every L, because the trajectory is a single exchangeable unit. This is the coverage face of the paper's thesis: the same mechanism drives the certification-count collapse of Section 8. Pre-registered, reported as produced (`e28`).

Severity: what each rung can detect. At ESS-like design noise (threshold SE 0.012, four-threshold core; E32), 80% power at the *net* rung needs a per-pair decline of 0.02–0.03 CDF points—realistic secular erosion—while the *persistent* rung needs 0.06–0.08, crisis-scale, and its power *falls* as the trajectory lengthens. Injecting known declines into the *real* ESS pairs, drawing each estimate from that country's own stratified-PSU bootstrap, puts the thresholds at ≈ 0.033 (net) and ≈ 0.075 (persistent), with realized size 0.001–0.007 against a nominal 0.10: the rule is markedly conservative, and non-certification is weak evidence of absence at *every* rung (E42). The powered instrument for secular decline is the net rung, which Section 8 co-headlines.

Validation, in three disclosed layers. *(i)* A sealed holdout (450 cells: ten DGP families $\times\ K \in \{25, \ldots, 320\} \times$ nine ρ; 800 reps) was run once—but an audit found its scorer evaluated the S2 studentized band rather than the deployed one, and its findings motivated the switch to U0 and the Theorem 5 architecture, so we label it development. *(ii)* A corrected-scorer rerun of the same grid scores the deployed band verbatim (fresh seed; the sealed config was not preserved—disclosed): all 450 cells floor-compatible, worst 0.882, deconvolution active only at $K = 320$. *(iii)* Because both layers had touched the final design, we sealed one further validation on six *new* DGP families, script hash recorded before first execution, reported as produced including a disclosed first-seal generator bug (E33): 120/120 floor-compatible, worst 0.878, zero activations below $K = 94$ as the floor requires.

Design-preserving stress test, honestly bounded. Subsampling the real AmericasBarometer, the deployed selector never fires; but the pooled $\widehat{\rho}$ turnover at deep subsampling coincides with ≈ 1 PSU per stratum, where the m-of-m bootstrap's variance estimate collapses—so the sweep demonstrates a failure mode of the $\widehat{\rho}$ diagnostic under sparse-PSU designs (coverage stayed at or above nominal throughout), not a saturation law; the regime characterization of Section 7 rests on the full-sample estimates instead (Supplementary Material).

7 Evidence from the ESS and the AmericasBarometer

The regime characterization. On real data the honest band reduces to clustered conformal—instructively. Across both surveys, both estimands, and three outcomes, cross-national inference is a uniformly *low design-noise* regime: $\widehat{\rho} \leq 0.22$ on LAPOP, at most 0.29 in the ESS subgroup scan, well below the $\rho_0 = 0.47$ cutoff. Two directional caveats attach to every $\widehat{\rho}$ we report: the PSU bootstrap resamples m-of-m without Rao–Wu–Yue rescaling (Rao et al., 1992) and single-PSU strata contribute no variance, so the estimates are downward-biased *floors*. The Rao–Wu–Yue rescaled rerun confirms the floors are tight: maxima 0.281 (ESS) and 0.222 (LAPOP), still severalfold below ρ_0, zero gate activations, every certification count unchanged; and singleton strata, which no rescaling repairs, are 1 of 7,028 in the core rounds.

The safe selector thus reduces to clustered PCB on all real data—not by assumption but by its own low-ρ gate—at the national unit, which the frontier below qualifies.

Why the deconvolution is unreachable, precisely.

Proposition 1 (Survey-scale unreachability)**.** *The deployed selector activates deconvolution only if a* need *gate* $\widehat{\rho}_{\mathrm{LCB}} > \rho_0$ *and a* reliability *gate* $D \le \tau_D$ *both hold,* $D = \max_t \widehat{\mathrm{SE}}(\widehat{s}_T^2)/\widehat{s}_T^2$. *Both are bounded by* algorithmic identities of the frozen procedure. (a) $\rho = \sqrt{\overline{v^2}}/s_{\mathrm{plug}} \in [0,1)$ *with* $s_{\mathrm{plug}}^2 = s_R^2 + \overline{v^2}$*: the need gate fires only when design noise is an appreciable fraction of the between-population signal.* (b) *Since* $\widehat{s}_T \le s_{\mathrm{plug}}$ *by construction,* $D \ge \sqrt{2/(K-1)}$*, so the reliability gate requires* $K \ge 1+2/\tau_D^2$*, i.e.* $K \ge 94$ *at the frozen* $\tau_D = 0.147$*. (Appendix A separates the deterministic statement from its kurtosis-dependent interpretation.)*

The two requirements pull apart on real surveys (Figure 2). On the ESS the noisiest subpopulations raise $\widehat{\rho}$ only to 0.29 while $K \le 33$ leaves $D \ge 0.25$: *neither* gate opens (42 cells, zero activations). The WVS is the mirror image—its full item-coverage sets ($K = 95$–105) make the reliability gate feasible, but the certification samples ($K = 59$–77) sit below 94, and in either denominator the weights-only $\widehat{\rho}_{\mathrm{LCB}} \le 0.09$ fails the need gate decisively. The survey large enough to estimate the deconvolution has almost no design noise to remove; the survey with real clustered noise has far too few countries.

The barrier belongs to the unit, not to surveys. Both gates are statements about how many exchangeable populations a survey contains and how much sample sits behind each, so refining the unit moves them together: K grows, relieving the floor $D \ge \sqrt{2/(K-1)}$, while the sample behind each population shrinks, raising ρ. ESS regions, with PSUs nested inside so each carries a genuine stratified-PSU bootstrap, let us trace that frontier on real survey data (`e49`, Figure 1); the estimand there is the small-area one this regime concerns, a region's departure from its own national distribution. Sweeping the minimum region size from 400 respondents to 25 takes K from 21–30 to 300–315 and $\widehat{\rho}_{\mathrm{LCB}}$ from 0.08 to 0.47. The reliability barrier clears cleanly: D falls below its gate in every round once units drop under 80 respondents, which is an algebraic consequence of K and replicates throughout. The need barrier only *approaches* its cutoff. Its maximum over the sweep is $\widehat{\rho}_{\mathrm{LCB}} = 0.476$ against $\rho_0 = 0.47$, in one of three rounds, at two of thirty-three unit-rounds—and that 1% margin is inside the specification noise of the estimand: accounting for the design variance of the *demeaned* curve rather than the raw one moves it to 0.460, below the cutoff, while dropping the countries that contribute a single region moves it to 0.495, above. Without demeaning at all the ratio never exceeds 0.24. What the frontier establishes is therefore the weaker and still useful claim: the reliability barrier is a property of K and dissolves at the subnational unit, while the need barrier comes within one percent of its cutoff on the small-area estimand and is not decisively crossed.

How coarse should the unit be? A feasibility floor, not an optimum. The same refinement that opens the gates also sets the band's width, and the direction is not obvious a priori: refining multiplies K, which tightens the conformal quantile, but shrinks the sample behind each unit, which inflates the design noise the band must carry. We tested it with the confounds removed (`e52`): within each country-round we build units of a *target* size by grouping whole PSUs at random, so size is controlled rather than filtered, composition is fixed, and each unit keeps its own stratified-PSU bootstrap. Over the range where the country set is constant (targets of 40 to 350 respondents), the half-width falls *monotonically* as units coarsen, 0.190 to 0.065: there is no interior optimum and refinement is strictly costly. An earlier sweep over NUTS regions appeared to show one; the Supplementary Material reports why it did not survive, because the artifact is easy to reproduce. What does bind is a corner. The conformal level is *infeasible* whenever $K < \alpha^{-1} - 1$: the required order statistic exceeds the sample and the band is $[0, 1]$ at any width. The practical rule is therefore the opposite of refinement—take the coarsest unit your question admits, subject to $K \geq \alpha^{-1} - 1$—and the cross-national default is on the right side of it ($K = 30$–34 against

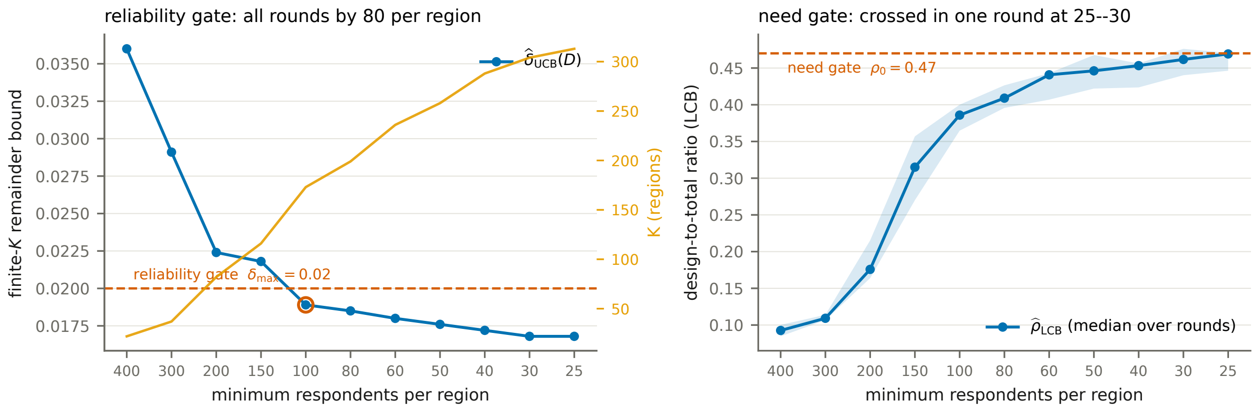

Figure 1: Refining the unit relieves both barriers (ESS regions, rounds 9–11; medians over rounds, shaded range). *Left:* the remainder bound falls past $\delta_{\max}$, in every round by 80 respondents per region. *Right:* $\widehat{\rho}_{\mathrm{LCB}}$ reaches ρ_0, crossing in one round at 25–30.

a floor of nine). This is a sharp and, as far as we know, unremarked constraint on cluster-conformal practice; it is a statement about the *calibration* unit of a transport band, and it does not bear on the unit of an estimand, which a research question fixes. Unit choice in the substantive sense—which areal partition to analyze—is the modifiable areal unit problem (Openshaw, 1984; Robinson, 1950), and we are not adding to it.

8 Political reanalysis: how many democracies really declined?

We analyze trust in parliament (`trstprl`) and satisfaction with democracy (`stfdem`) in ESS rounds 9–11 (2018–2024, the window whose integrated files ship PSU/stratum identifiers), then extend every trajectory to the full record (rounds 1–11, 2002–2024), asking at each rung how many countries survive. The comparative literature has grown skeptical of a continent-wide trust decline (Norris, 2017; Voeten, 2017; Wuttke et al., 2022; Valgarðsson et al., 2025); we add a formal instrument that says so and the wrong-unit diagnosis of *why* the over-reading arises. The dominant global trend estimates (Valgarðsson et al., 2025) rest on latent-trait models whose intervals carry no design-based uncertainty layer (Tai et al., 2024) and whose

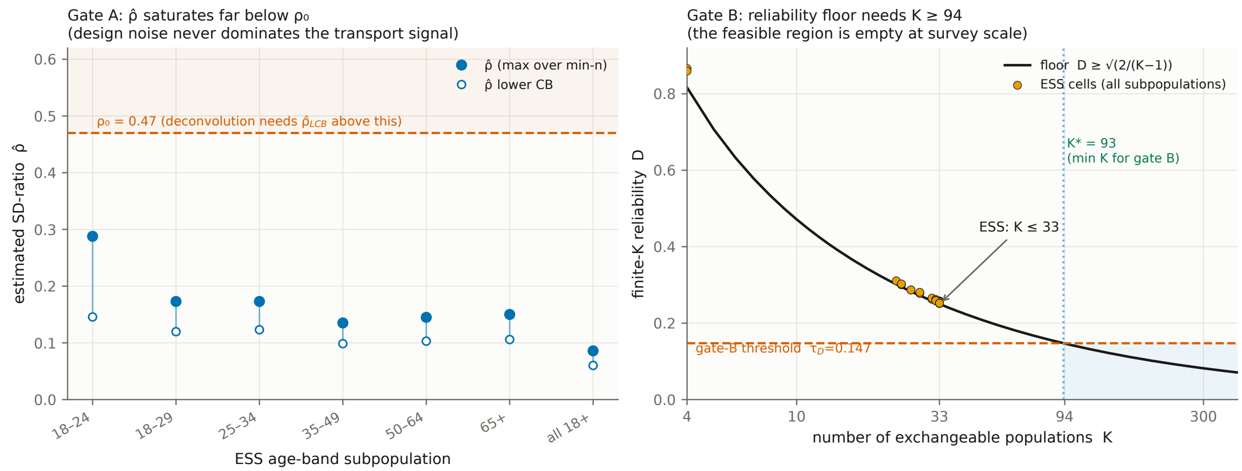

Figure 2: The two barriers at the national unit. *Left:* ESS age-band subpopulations (42 cells), $\widehat{\rho}$ vs ρ_0. *Right:* D vs its floor $\sqrt{2/(K-1)}$; only WVS full item-coverage sets reach $K \geq 94$, and there the need gate fails.

data end in 2019, where our window begins.

A hierarchy of claims, and one band beneath it. The claims differ in their unit. A *pairwise* claim asks whether some pair of waves shows a distributional shift; *any-pair* aggregates these within a country; *net* asks whether the first-to-last span declined; *persistent country-wide* asks whether the whole distribution declined *simultaneously* across the trajectory; *Bonferroni* further controls the number of countries examined. Each rung is strictly stronger than the last, and conflating them is exactly the over-detection this paper is about.

The rungs are not separate tests: each of them, every sub-span, and the reverse direction is a deterministic function of one object—the country's surface of wave-pair contrasts—so all are certified on one coverage event.

Proposition 2 (Claim-family transfer)**.** *Let $\widehat{B}$ be a simultaneous $1-\alpha$ band for a country's contrast surface $\theta = \{\theta_{a,b}(t)\}$ over every ordered pair of observed rounds, and let $\{A_j\}$ be* any *family of claims, each a set of surfaces. Define the certified subfamily $J = \{j : \widehat{B} \subseteq A_j\}$. Then* $\Pr\left(\theta \in A_j \text{ for every } j \in J\right) \geq 1-\alpha$*, with no dependence on $|J|$: on $\{\theta \in \widehat{B}\}$, set inclusion delivers every certified claim at once.*

The proof is the line just given; the consequence is structural. Reading all rungs off a single studentized sup-t band over the whole contrast family—one critical value, both directions—removes the corrections a per-rung construction needs: no Bonferroni across a country's pairs, no separate budget for certified recoveries, and no endpoint-selection cost, since every choice of first and last round is certified at once. The price is a wider critical value, and we report what it costs.

Which instrument certifies what. Within-country *certification* is a design-based simultaneous (sup-t) band on wave-pair CDF differences from the stratified-PSU bootstrap (Rao and Wu, 1988; Wolter, 2007), an asymptotic guarantee in which exchangeability plays no role; the conformal *transport* band of Theorems 3–5 calibrates cross-country prediction and the regime diagnostics. Headline counts are certification counts, and the theory sections' finite-sample language does not transfer to them.

The counts collapse up the hierarchy. For `trstprl` over rounds 9–11, a marginal pairwise reading flags 20 of 30 countries; an any-pair design-aware band drops this to 12; net first-to-last decline holds in 6 (Austria, Belgium, Estonia, the United Kingdom, Greece, the Netherlands); and the *persistent, design-aware simultaneous* band over the preregistered low-trust core—the strongest rung—certifies **one**, Greece, which also survives Bonferroni across countries. One disclosure is essential: *Greece fielded only rounds 10 and 11 here*, so its trajectory is a single pair and the top rungs coincide for it (as for Czechia and Israel, which do not certify). Among the 27 two-pair countries *none* certifies persistence, while the United Kingdom, Belgium, and the Netherlands pass the *plug-in* persistent rung and are demoted by the design-aware layer alone. The honest co-headline: **net decline in six of thirty; persistent decline in one, on a single pair**. The severity analysis (Section 6) explains why: the persistent rung has 80% power only against crisis-scale declines, so its non-certifications are weak evidence of absence. For `stfdem` the pattern repeats (23 marginal, 14 any-pair, 8 net, one persistent—Greece, again one pair, failing Bonferroni there). Greece is the case the literature reads as a crisis-era legitimacy collapse (Armingeon and Guthmann, 2014; Torcal, 2014), though those accounts concern 2009–2015 while the certified pair is 2020/21→2023/24.

The long window: the secular question, asked properly. Extending every trajectory to its full record (rounds 1–11; median six pairs, up to ten; rounds 1–8 weights-only) reframes the twenty-year narrative in both directions (`e36`). *Zero* of the thirty-three countries with three or more usable rounds certify persistent distribution-wide decline over their full record, plug-in included. The null is not a bare one: reversing the inequality certifies *recoveries*, and across the 1,173 ordered contrasts of the trust series certified movement is directionally balanced (193 declining against 212 rising). Inside the same band, and therefore at the same α for that country, *twenty-three* of thirty-three countries certify at least one decline *and* at least one recovery, on each outcome separately and twenty on both (`e50`). The two directions share one critical value by construction, so no within-country correction is needed; the count *across* countries is a conjunction of thirty-three separate coverage events and carries no familywise control, as do all the country counts in this section unless the Bonferroni rung is named. Whatever European trust did over two decades, it was not a monotone slide. We stop short of calling it episodic: that would additionally require same-signed runs and returns toward baseline, which these counts do not establish.

Under Proposition 2—one band, every ordered contrast and both directions at $\alpha = 0.10$ jointly (`e50`)—net erosion over the whole span certifies in *eight* countries (Cyprus, Spain, the United Kingdom, Greece, Hungary, Israel, Italy, Ukraine; Table 2). Against the per-rung construction the joint statement costs France at the span rung ($9 \to 8$) and six countries at the any-pair rung ($28 \to 22$). The two are not otherwise comparable: the per-rung run treats only consecutive ESS rounds as adjacent, the joint band consecutive *observed* rounds, so for a country that skipped waves the joint "persistent" rung asks a weaker question—every consecutively observed pair, not every two-year step. We report the persistent zero under

Table 2: Certified net decline in parliamentary trust, 2002–2024, read off one band covering every ordered contrast and both directions at $\alpha = 0.10$ jointly. *Decline* $\geq$: simultaneous lower bound over the preregistered core, in CDF points. *Contrasts*: ordered round pairs certified declining, of all admissible; their share is the endpoint-free erosion index. The certified set is unchanged at any weights-only design effect in $[0.6, 2.0]$.

country	rounds	decline $\geq$	contrasts	share
Ukraine	2–11	25.9	7/15	0.47
Cyprus	3–11	6.2	13/21	0.62
United Kingdom	1–11	6.1	11/55	0.20
Greece	1–11	3.1	8/15	0.53
Israel	1–11	2.6	10/28	0.36
Italy	1–11	1.6	5/15	0.33
Hungary	1–11	1.1	9/55	0.16
Spain	1–11	0.6	32/55	0.58

both definitions, and it is zero under both. The under-detection face of the wrong unit shows in the ledger of contrasts: Hungary certifies its 22-year span while only 9 of its 55 ordered contrasts certify, and the United Kingdom 11 of 55—erosion that a reading built from adjacent waves alone would mostly miss. Because every choice of first and last round is certified at the same time, the band also delivers the *share of a country's ordered contrasts that certify a decline*, a within-country summary no endpoint selects. It is a lower bound rather than an estimate, and it is not comparable across countries: the shared critical value grows with the length of the record (correlation 0.96 with the number of rounds) while the denominator grows quadratically, so a country observed eleven times is penalized twice over against one observed six times. Within a country it is informative—Spain certifies over 32 of its 55 contrasts and Slovenia over 21 of 55, neither entering the eight because their span contrast happens not to certify, which is precisely the endpoint sensitivity the joint band exposes. Ranked across countries it would be an artifact of record length, and we do not rank it. Ukraine's certified span decline of *at least* 25 points likewise runs from the post-Orange-Revolution wave of 2005 to a wartime sample: a certified statement about those two waves.

The reclassified countries are the weak-signal, higher-noise cases: *certified by the marginal*

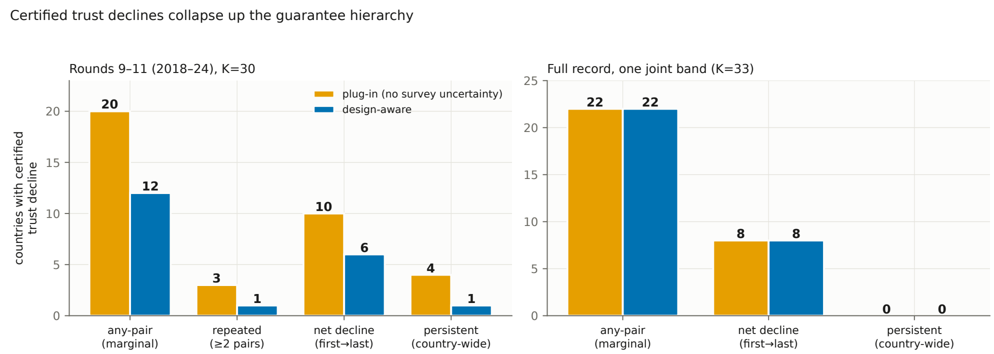

Figure 3: Certified declines collapse up the hierarchy ($\alpha = 0.10$). *Left*, rounds 9–11 ($K = 30$), plug-in and design-aware bars. *Right*, the full record read off one joint band ($K = 33$), where the two bars coincide by construction.

plug-in but inconclusive under the design-aware band—never "false positives," a term we reserve for simulation with known truth.

Two confounds this window cannot exclude. Round-10 fieldwork coincided with the pandemic's rally-round-the-flag movements, so "persistent decline through this window" is a claim made against a rally; and nine countries switched to self-completion at round 10, confounding wave-pair shifts with mode effects—a bias no design bootstrap absorbs. A mode audit built from the data's own mode variable bounds it: the Greek pair is mode-constant and net counts are unchanged when the switchers' confounded pairs are excluded, only their any-pair claims being affected.

Robustness across age groups. A preregistered age-group analysis confirms the result is not an artifact of pooling ages; youth certify zero persistent declines as an underpowered null, not youth stability (Supplementary Material).

A second, global target: Foa–Mounk deconsolidation. We preregistered a reanalysis of the "democratic deconsolidation" thesis (Foa and Mounk, 2016, 2017) on the WVS/EVS trends (1981–2022, 59–77 countries with ≥ 2 waves), recoding the five-item battery so that deconsolidation is a persistent decline over each scale's preregistered core. The verdict is calibration, not denial—but the surviving phenomenon is not the one the thesis was about. A *marginal wave-pair reading of the same data*—a criterion we construct to instantiate the bottom rung, since Foa and Mounk's own empirics are cohort comparisons—flags 36–64 countries per item against 6–17 (8–24%) at the persistent rung. That 2.6–6.5$\times$ ratio compounds two changes, and we separate them: holding the inference layer fixed, the *rung* alone accounts for 1.7–4.8$\times$ (plug-in) or 1.9–4.8$\times$ (design-aware), and propagating design uncertainty at fixed rung accounts for the remainder. The unit is the larger effect, but it is not the whole of the headline number. Three caveats. The WVS ships no PSU/stratum identifiers, so the weights-only bootstrap understates design variance, which can only *inflate* counts: the deff rerun confirms the gap is a lower bound—at variance $\times$1.5 and $\times$2.0 the certified core holds at 13 and 12 countries and the gap widens to 3.0–8.3$\times$. The per-item lists carry no across-country familywise control, so we aggregate to multi-item co-certification before interpreting geography. And the certification sets sit below the $K \geq 94$ bound, so every WVS band uses the full-α anchors of Theorem 5. The youth "embrace authoritarianism" claim is only half-supported (Supplementary Material).

The certified core: where the surviving deconsolidation lives. Aggregating per-item certifications yields a positive characterization (Figure 4; E30): of 38 countries certified on ≥ 1 item, 13 form a *certified core* on ≥ 2, headed by Albania on four and Bosnia, Ecuador and Tunisia on three. The core is not the consolidated West but post-communist and Arab-Spring states, with fragile Latin American and African cases alongside—not the mature democracies the thesis (Foa and Mounk, 2016, 2017) concerned—and it formalizes, with per-country error control, the decline-from-euphoric-baselines pattern the post-communist disillusionment literature described qualitatively (Mishler and Rose, 1997; Inglehart and Catterberg, 2003). Three qualifications travel with it. Five core members are electoral autocracies in the V-Dem classification, where falling stated support reflects autocratic legitimation rather than deconsolidation in the consolidated-democracy sense. The West enters the core exactly twice (Finland and Switzerland, on the strongman/army pair, which we flag as a probable administration artifact), with single-item entries elsewhere: no consolidated Western democracy certifies on support for a democratic system—consistent with Wuttke et al. (2022), though not that the West is absent. And the geography is partly window-contingent: recertifying on fieldwork ending by 2017 (`e46`) keeps eight of the thirteen but drops all three Arab-Spring cases, so the post-communist component is window-stable and the Arab-Spring component rests on WVS wave 7. A further objection deserves a direct answer: certified attitudes do not forecast institutions—core membership predicts subsequent V-Dem electoral-democracy decline no better than the marginal flag ($n = 13$ versus 54; Fisher $p = 0.89$), too small a comparison to establish decoupling, which recent panel work anyway expects (Ouattara and van der Meer, 2023; van der Meer and Janssen, 2025). The estimand is the attitude trajectory itself: certification is descriptive error control, not a forecast. Item-level caveats, and the narrower-than-reputation reading of Turkey, the Philippines and Mexico, are in the Supplementary Material.

9 Discussion

Uncertainty in claims about repeated cross-national surveys is usually computed for the wrong unit—twice. We gave a multiplicity-honest simultaneous band for the claim unit and a hard boundary for the calibration unit: non-identified without the design-noise law (Theorem 1), unreachable at cross-national scale with it (Proposition 1). The honest deployed object is the reduction—a clustered band exact at any K inside a selection-free selector (Theorems 3, 5)—and the design-aware layer certifies, rather than corrects, at survey scale.

When it reduces to ordinary conformal. At the national unit the method reduces *exactly* to clustered population conformal: the machinery certifies that the simple band is the honest one and reports the ρ that justifies it. The deconvolution regime needs many exchangeable units *and* comparable design noise—never both across countries, but reachable one unit down (Section 7) and routine in simulation at many-unit scale (E31: $K \approx 220$–300 small areas, deconvolution on up to 98% of draws at 0.69–0.74× the conservative width with valid coverage).

Limitations. Four scope conditions, stated fully in the Supplementary Material. *Sampling error only:* the machinery is silent on measurement non-invariance (wording and translation changes, EVS/WVS house effects, ESS round-10 mode switches), which produces exactly the shifts the bands certify; certifications straddling such changes are conditional on measurement constancy. *Variance-estimator bias:* the deployed m-of-m PSU bootstrap (Rao et al., 1992; Wolter, 2007) biases design variance downward; the Rao–Wu–Yue rescaled rerun leaves every certification count and gate decision unchanged (Supplementary Material), though single-PSU strata still carry no variance information. *A superpopulation postulate:* transport-band exchangeability is over a non-sampled, panel-attriting country set (the certifications do not use it), and holds worse for the post-communist bloc than elsewhere. *Metadata:* without design metadata (WVS) only weights-only replication is possible, and the Gaussian-calibrated finite-K remainder is confined by Theorem 5 to the $K \geq 94$ regime.

The general lesson. Attaching uncertainty to the right unit changes substantive conclusions: persistence in none of thirty-four European democracies over two decades yet span erosion in eight, mostly invisible to adjacent-wave readings; a several-fold over-count of persistent deconsolidation, with the surviving core concentrated outside the consolidated West. The pattern is not confined to surveys: wherever prediction is calibrated on estimated objects—small-area model outputs, learned features, meta-analytic effects (Remark 1)—the calibration targets carry their own error, and the claim unit is often coarser than the trend asserted. The right response is not to widen every band reflexively but to state the strongest object the data support, correct the estimation noise when it can be identified, and abstain honestly when it cannot.

A Proofs and efficiency results

Full statements and proofs of all theorems, the efficiency results, and the modulation self-inclusion deficit are in Section S1 of the supplementary material; each theorem is paired with a machine-checked contract test.

B Frozen selector specification

The frozen constants, their development-grid derivation, and the validation protocols are in the replication package.

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Competing Interests

Competing interests: the author(s) declare none.

Data and Code Availability

All simulation, theory, and benchmark results are deterministic, reproducing exactly under fixed seeds with no external data, and each theorem is paired with a machine-checked contract test. The ESS, World Values Survey, and AmericasBarometer/LAPOP microdata are licensed to their providers and not redistributed; the replication package documents the exact variables, waves, and retrieval steps. The full package—the `pcb` module, all simulations and contract tests, and the analysis pipeline—will be deposited in the Political Analysis Dataverse (Harvard Dataverse) upon conditional acceptance.

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Figure 4: Co-certification across the five battery items (WVS/EVS 1981–2022; $K = 59$–77 per item). Countries above the dashed line form the ≥ 2-item certified core.